llmXive Automated Review of LatentSkill: From In-Context Textual Skills to In-Weight Latent Skills for LLM Agents

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a strong empirical case for LatentSkill, with most claims supported by the provided tables and appendices. However, a few specific numerical claims in the text do not align perfectly with the data presented in the tables, requiring minor corrections to ensure factual accuracy.

First, in Section 4.5 (“Composable”), the text asserts that Component Merging adds “2 on the unseen split” relative to Look-Only. According to Table 4 (Skill composition results), Look-Only achieves 13/18 successes on the unseen split, while Component Merging achieves 14/18. This represents a gain of exactly 1 episode, not 2. This discrepancy should be corrected to “1 episode” to maintain consistency with the reported data.

Second, in Section 4.4 (“Controllable”), the text discusses the performance loss of applying a global $\alpha = 0.6$ to unseen tasks. While the specific point losses cited for Heat (21.74), Pick2 (17.65), and Clean (12.90) are mathematically correct based on Table 5, the text frames this around the “seen-split global optimum $\alpha = 0.6$“. However, Table 5 shows that the *overall* average success rate on the unseen split actually peaks at $\alpha = 0.5$ (70.90%), not $\alpha = 0.6$ (69.40%). The text should clarify that the comparison is specifically against the *seen-split* optimum being applied to the unseen split, rather than implying $\alpha = 0.6$ is the global optimum for the unseen domain.

Finally, in Section 4.3 (“Structured”), the text claims that OOD skills form “separated clusters” and provides specific within-domain similarity scores (0.783, 0.9664, 0.9681). To fully support the claim of separation, the text should also explicitly state the cross-domain similarity values or the inter-cluster distances for these OOD groups. Without these comparative figures, the claim that they are “separated” relies on the reader inferring the gap, which is less rigorous than the explicit reporting used for the in-domain clusters.

These are minor textual inaccuracies that do not undermine the core scientific findings but should be fixed to ensure the manuscript’s precision.

Action items.

- [**writing**] Section 4.3 claims OOD skills form ‘separated clusters’ but omits cross-domain similarity values needed to verify separation. Explicitly report cross-domain metrics to support the claim.
- [**writing**] Section 4.4 cites a global optimum of =0.6, but Table 5 shows the unseen average peaks at =0.5 (70.90%). Clarify that 0.6 is the seen-split optimum applied to unseen tasks to avoid confusion.
- [**writing**] Section 4.5 states Component Merging adds ‘2’ unseen episodes over Look-Only. Table 4 shows 14/18 vs 13/18, a gain of only 1. Correct the text to match the table data.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear and well-constructed bar chart that effectively supports the caption's claims. The axes are labeled with units, the legend distinguishes between Pretrain and SFT conditions, and the data visualization aligns perfectly with the textual description of attn_o and mlp_down exhibiting higher gaps.

Figure 2

The figure effectively visualizes the clustering of skills, but the axis scaling notation in the caption conflicts with the raw tick labels, and the left panel's legend is incomplete regarding the distinction between hollow and filled markers.

Figure 3

The figure effectively displays scale-performance curves with clear axes and markers, but the top panel legend includes an unused entry ('Unseen') and the caption contains a formatting error regarding the variable name for the x-axis.

Figure 4

The figure effectively visualizes the contrast between in-context and in-weight skills, but the caption contains a grammatical error omitting the subject and fails to describe key visual components like the context budget and exposure warnings.

Figure 5

The figure provides a clear and well-structured visual overview of the proposed pipeline, effectively illustrating the three main stages. However, the caption contains a significant grammatical error where the subject of the overview is missing.

Action items.

- **[science]** Figure 2: The caption states 'Axes are scaled by $10^{-}2$', but the axis tick labels (-6 to 4) appear to be the raw values. If the data is scaled, the axis labels should reflect the actual values (e.g., -0.06 to 0.04) or explicitly state 'x10^-2' to avoid misinterpretation of the magnitude.

- [**writing**] Figure 2: The legend in the left panel defines 'ALFWorld' and 'Search' with solid square colors, but the plot uses hollow squares for 'Pretrain' and filled squares for 'SFT'. The legend fails to map the specific shapes (hollow vs. filled) to the training stages, creating ambiguity about which points correspond to which condition.
- [**science**] Figure 3: The top panel legend defines 'Seen' and 'Unseen' line styles, but the caption states the bottom panel is on the 'unseen split' while the top panel does not specify the split. If the top panel is also unseen, the legend is redundant; if mixed, the caption is incomplete. Additionally, the top panel legend defines 'Unseen' (solid gray) but no solid gray line is plotted, only a dashed gray line for 'Seen', creating a legend mismatch.
- [**writing**] Figure 3: The caption contains a formatting error where the variable for the LoRA injection coefficient is missing (e.g., 'coefficient $$' instead of 'coefficient α'), making the text technically unreadable.
- [**writing**] Figure 4: The caption reads 'The key advantages of over in-context skill', which is grammatically incomplete and missing the subject (likely 'LatentSkill' or 'in-weight skills') that the figure illustrates.
- [**science**] Figure 4: The diagram includes a 'Context Budget' bar at 98% and a 'Plaintext Skill Exposure' warning, but the caption fails to mention these specific visual elements or explain their relevance to the claimed advantages.
- [**writing**] Figure 5 caption: The phrase 'Overview of .' is grammatically incomplete and missing the subject (likely 'LatentSkill' or 'the framework').
- [**writing**] Figure 5 caption: The text 'Overview of .' repeats the same grammatical error found in the main caption body.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology that, while standard in specific subfields of LLM adaptation, creates barriers for a broader audience. The most significant issue is the use of the acronym "OOD" (Out-of-Distribution) in Section 4.3 ("Structured: Semantic Geometry of the LoRA Weight Space") without prior definition in the main text. While defined in the appendix, the acronym appears in the body before the reader encounters the definition, violating standard clarity practices.

The phrase "in-weight latent skills" in the title and abstract is awkward and non-standard. "In-weight" is not a common prepositional phrase in this context; "weight-embedded" or "parameter-embedded" would be more precise and accessible. Similarly, the term "skill compiler" (Section 3) is potentially misleading. In computer science, a compiler implies a deterministic translation process, whereas the paper describes a learned hypernetwork. “Skill encoder” or “adapter generator” would better reflect the mechanism and reduce cognitive load for readers unfamiliar with the specific “compiler” metaphor used here.

In Section 4.5 and Appendix B.1, the authors refer to “mistakes components.” This is ambiguous. Does it refer to a component that generates mistakes, a component that prevents them, or a component trained on error data? Replacing this with “error-avoidance components” or “negative constraint modules” would clarify the function without requiring the reader to infer meaning from context.

Finally, the “Low-Rank Encoding Analysis” in Appendix C introduces “stable rank” without a brief definition or intuition. While experts in matrix analysis know this term, a general reader in AI may not. A single sentence explaining that it measures the effective dimensionality of the weight matrix would make this section accessible to a wider audience. These changes are minor but essential for ensuring the paper’s contributions are understood by non-specialists.

Action items.

- [**writing**] Define ‘OOD’ (Out-of-Distribution) at its first occurrence in the main text (Section 4.3) rather than assuming reader familiarity, as the acronym is used repeatedly before definition in the body.
- [**writing**] Replace the phrase ‘in-weight latent skills’ in the title and abstract with ‘weight-embedded skills’ or ‘parameter-embedded skills’ to avoid the non-standard and slightly awkward ‘in-weight’ construction.
- [**writing**] Clarify the term ‘mistakes components’ in Section 4.5 and Appendix B.1. It is unclear if this refers to error-handling logic, negative constraints, or a specific module name; use a plainer description like ‘error-avoidance components’ or define it explicitly.
- [**writing**] Replace ‘skill compiler’ with ‘skill encoder’ or ‘adapter generator’ in Section 3. ‘Compiler’ implies a static translation process, whereas the text describes a learned hypernetwork; the current term may confuse readers expecting a traditional compiler.
- [**writing**] Define ‘stable rank’ in the context of the Low-Rank Encoding Analysis (Appendix C) for non-specialist readers, as it is a specific linear algebra metric not universally known outside the field.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The logical consistency of the paper is generally strong, with clear causal chains established between the proposed method (LatentSkill) and the observed benefits (efficiency, performance, robustness). The argument that moving skills from context to weights reduces token overhead and improves robustness to prompt injection is well-supported by the experimental design comparing In-Context Skill vs. LatentSkill.

However, there are specific instances where the textual claims do not perfectly align with the provided data tables, creating minor logical gaps:

1. **Numerical Discrepancy in Section 4.3:** The claim that “four of the six tasks share the same optimal α“ (Section 4.3, paragraph 2) appears inconsistent with Table 10 (Appendix A.5). A direct comparison of the “Seen” and “Unseen” optimal α values in Table 10 reveals that only “Heat” (0.3/0.3) and “Cool” (0.6/0.6) share the exact same optimal value. “Pick” (0.5/0.6), “Look” (0.6/0.5), “Clean” (0.3/0.8), and “Pick2” (0.6/0.8) all differ. This overstatement weakens the logical support for the claim that the injection range is uniformly robust across tasks and splits.
2. **Geometric Interpretation in Section 4.3:** The text states that after SFT, “inter-cluster distance decreases... while both within-domain and cross-domain similarities increase.” While mathematically possible (clusters shrink and move closer), the causal explanation that “SFT introduces shared agent-level behavioral patterns” leading to this specific geometric contraction needs a slightly more explicit logical bridge. The current phrasing risks a slight ambiguity: does the increased similarity refer to the *average* similarity (which would naturally rise if clusters shrink) or the *centroid* similarity? Clarifying that the *intra-cluster* variance reduction drives the similarity increase, while the *inter-cluster* shift is driven by the shared patterns, would tighten the logic.
3. **Definition of “No Interference” in Section 4.5:** The paper defines composability success as preserving the target skill’s capability. The evidence provided is binary success rates (e.g., “losing none of Look-Only’s original successes”). Logically, “no interference” could also imply maintaining the *quality* or *efficiency* of those successes. If the Component Merged model takes significantly more steps to complete the Look-only tasks it successfully solves, a subtle form of interference exists. While the binary metric supports the primary claim, acknowledging the scope of “no interference” (i.e., strictly success rate vs. full trajectory quality) would prevent over-interpretation.

These issues are primarily matters of precise data reporting and scope definition rather than fundamental flaws in the research logic. The core causal mechanisms (hypernetwork generation, LoRA composition) are sound and well-argued.

Action items.

- **[writing]** Numerical Discrepancy in Section 4.3: The claim that “four of the six tasks share the same optimal α“ (Section 4.3, paragraph 2) appears inconsistent with Table 10 (Appendix A.5). A direct comparison of the “Seen” and “Unseen” optimal α values in Table 10 reveals that only “Heat” (0.3/0.3) and “Cool” (0.6/0.6) share the exact same optimal value. “Pick” (0.5/0.6), “Look” (0.6/0.5), “Clean” (0.3/0.8), and “Pick2” (0.6/0.8) all differ. This overstatement weakens the logical support fo
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Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the generalizability and security implications of LatentSkill that slightly exceed the empirical evidence provided.

First, the abstract and introduction claim that weight-space skills are "less exposed" as plaintext. While the "Extract" attack results (Table 5) show that the model does not verbatim reproduce the skill text when prompted, this does not equate to the weights being secure or "less exposed" in a broader security sense. The weights themselves encode the skill, and without analysis of weight extraction resistance (e.g., against model inversion or membership inference attacks), the claim overstates the security benefit. The limitation section correctly notes this, but the main text should be more precise, framing it as "reducing direct plaintext exposure in the prompt" rather than implying a robust security guarantee.

Second, the conclusion states that "latent skill weights offer a practical substrate for building LLM agents." This is a broad generalization based on experiments limited to two benchmarks (ALFWorld, Search-QA), a single backbone model (Qwen3-8B), and a fixed LoRA configuration. The paper does not evaluate complex, long-running agents, multi-agent collaboration, or open-ended environments. While the limitations section acknowledges these gaps, the conclusion's phrasing suggests a level of maturity and generality that the current data does not fully support.

Finally, the claim that the hypernetwork "spontaneously learns a semantically structured weight space" (Section 4.3) is slightly over-interpreted. The observed clustering in the MDS visualization could be a direct result of the distinct skill documents in the training set rather than an intrinsic, universal property of the hypernetwork's weight space. The paper should clarify that this structure is observed under the specific training regime and data distribution used, rather than implying it is an inherent feature of all such hypernetworks.

Action items.

- [**writing**] The claim that LatentSkill is 'less exposed' (Abstract) overstates security. 'Extract' attacks only show text isn't regurgitated, not that weights are secure against inversion. Temper to 'reduces direct plaintext exposure in the prompt'.
- [**writing**] The conclusion that weights offer a 'practical substrate' (Conclusion) overreaches. Results are limited to two benchmarks and one backbone. Qualify claims to reflect untested complexity in real-world agent workflows.
- [**writing**] Claiming the hypernetwork 'spontaneously learns' structure (Sec 4.3) over-interprets MDS. Clustering may stem from training data distribution, not intrinsic weight properties. Clarify this is specific to the training regime.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper addresses safety and ethics primarily through the lens of prompt injection and skill extraction, arguing that moving skills from context space to weight space (LoRA) mitigates these risks. While the motivation is sound, the empirical evidence provided to support these security claims is currently insufficient and lacks necessary rigor.

In Section 5 ("Sensitivity and Security"), the authors evaluate robustness against "Hijack" and "Extract" attacks. However, the description of the "Hijack" attack is vague, referring only to a "malicious system-override instruction." To validate the claim that LatentSkill is robust, the specific prompt used must be disclosed, and it should be demonstrated that this prompt represents a realistic, high-severity threat (e.g., a sophisticated indirect injection as described in Greshake et al., 2023) rather than a trivial override that might fail against any model. Furthermore, the "Extract" attack claims that weight-space skills are "substantially harder" to recover. The paper provides no quantitative metric for this difficulty (e.g., the success rate of a specific extraction attack, the perplexity of the recovered text, or the number of queries required). Without such metrics, the assertion that the method improves security remains a qualitative hypothesis rather than a proven result.

Regarding data ethics, Appendix A ("Training Details") states that the pretraining corpus consists of "approximately 171K deduplicated skill documents crawled from GitHub." The authors do not mention any filtering for software licenses (e.g., ensuring no proprietary or non-permissive licenses were included) or any process for handling potentially sensitive or private information that might exist in public repositories. Given the nature of "skills" which could encode proprietary business logic or sensitive procedures, a statement on data provenance, license compliance, and privacy risk assessment is required.

Finally, the "Limitations" section correctly notes that weight-space storage "does not by itself provide a complete security guarantee" and that adapters can still be vulnerable to poisoning or model-access attacks. This critical nuance is currently buried in the limitations and risks being overlooked by readers who may interpret the "Sensitivity and Security" results as a definitive solution. This caveat should be elevated to the main security discussion to ensure a balanced and accurate representation of the method's safety profile.

Action items.

- [**writing**] The 'Hijack' attack evaluation (Section 5) uses a generic 'malicious system-override' instruction. To substantiate the claim of improved robustness, the authors must specify the exact prompt used and demonstrate that it is a realistic, high-severity attack vector (e.g., referencing Greshake et al. 2023) rather than a trivial override.
- [**writing**] The 'Extract' attack (Section 5) claims weight-space skills are harder to recover. The paper lacks a quantitative measure of this difficulty (e.g., perplexity of extracted text or success rate of a specific extraction attack). Without this, the security claim remains qualitative and potentially overstated.
- [**writing**] The pretraining data is crawled from GitHub (Appendix A). The authors must explicitly state whether they filtered for licenses (e.g., excluding proprietary code) and whether they obtained consent or performed a risk assessment for using potentially sensitive or private skill documents in a public model.
- [**writing**] The 'Limitations' section (Section 6) acknowledges that weight-space skills do not provide a 'complete security guarantee' against adversaries with model access. This crucial caveat should be moved to the main 'Sensitivity and Security' section to prevent readers from misinterpreting the method as a definitive security solution.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling empirical case for LatentSkill, with robust main results on ALFWorld and Search-QA benchmarks. The performance gains over in-context baselines are substantial and consistent across splits. However, the scientific evidence supporting the specific claims of "structured," "controllable," and "composable" properties requires stronger statistical grounding.

First, the sensitivity analysis in Appendix app:perturbation (Table tab:sensitivity_full_table) reports point estimates for performance under perturbations but omits measures of variance or statistical significance. Given the sample sizes (140/134 episodes for ALFWorld, 500 for Search-QA), the authors should compute 95% confidence intervals or perform paired t-tests (or non-parametric equivalents) to confirm that the observed margins (e.g., +21.4% on ALFWorld seen) are not artifacts of random variance. Without this, the robustness claims remain descriptive rather than statistically validated.

Second, the claim of "composability" is supported by a single experiment involving the Look and Pick skills across 31 episodes (Table tab:compose). While the case studies in Appendix app:case_study provide valuable mechanistic insight, a sample of 31 episodes is too small to generalize the finding that "skills can be composed" as a broad property of the framework. The failure modes of Direct and Text merging are well-documented, but the success of Component Merging on this specific pair does not prove it will hold for arbitrary skill combinations. The authors should either expand the composition experiments to include at least 3-4 distinct skill pairs or explicitly limit the claim to "demonstrated feasibility on complementary skill pairs."

Third, the evidence for "structured" weight space relies heavily on visual inspection of MDS plots (Figure fig:lora_mds). While the clusters appear distinct, the paper lacks quantitative metrics of cluster separation (e.g., silhouette coefficients, Davies-Bouldin index, or permutation tests for cluster significance). The reported cosine similarities (0.982 vs 0.910) are informative but do not account for the high dimensionality of the weight space or the potential for overfitting in the MDS projection. A statistical test confirming that the observed inter-cluster distances are significantly larger than expected by chance would strengthen this claim.

Finally, the injection coefficient analysis (Table tab:scale) shows a clear inverted-U curve, but the optimal α values vary significantly across tasks (e.g., 0.3 for Heat vs 0.8 for Pick2). The paper notes this variation but does not provide a statistical analysis of the stability of these optima across different random seeds or data splits. Reporting the variance of optimal α across multiple runs would clarify whether the "controllable" property is robust or highly sensitive to specific task instances.

Overall, the core performance claims are well-supported, but the auxiliary claims regarding the geometric and compositional properties of the latent space require more rigorous statistical validation to be considered scientifically robust.

Action items.

- [**science**] The sensitivity analysis (Appendix app:perturbation) lacks statistical significance testing. With N=140/134 episodes per ALFWorld split and N=500 per Search-QA dataset, the authors should report confidence intervals or p-values for the observed performance gaps between LatentSkill and In-context baselines to rule out variance-driven claims.

- [**science**] The compositionality claim relies on a single skill pair (Look/Pick) across 31 episodes (Table tab:compose). This sample size is insufficient to generalize the 'composable' property to arbitrary skills. The authors should either expand the composition experiments to more skill pairs or temper the claim to 'demonstrated on a specific complementary pair'.

- [**science**] The OOD generalization analysis (Section 4.3) uses MDS on 42 total skills (8 in-domain + 34 OOD) without reporting statistical measures of cluster separation (e.g., silhouette scores or permutation tests). Visual separation in Figure fig:lora_mds is suggestive but not rigorous evidence of structured geometry.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the paper is generally descriptive, relying heavily on point estimates (success rates, exact match scores) without accompanying measures of uncertainty. While the magnitude of the reported improvements (e.g., +21.4% on ALFWorld seen split) appears substantial, the lack of statistical rigor in the sensitivity and compositionality analyses weakens the empirical support for the claims of robustness and composability.

Specifically, the sensitivity analysis in Appendix \ref{app:perturbation} presents results from what appears to be a single evaluation run per perturbation type. For stochastic LLM agents, performance can vary significantly between runs due to sampling variance. Reporting single-point estimates without standard deviations, standard errors, or confidence intervals makes it impossible to determine if the observed differences between \method{} and the in-context baseline under perturbations (e.g., the 3.6 point drop for \method{} vs 3.1 for in-context) are statistically significant or merely noise. The authors should re-run these experiments with multiple random seeds (at least 3-5) and report the mean and variance.

Similarly, the analysis of the injection coefficient α (Appendix \ref{app:scale}) identifies optimal values based on peak performance but does not statistically validate the "inverted-U" shape or the stability of the optimal range. Without statistical tests (e.g., repeated measures ANOVA or pairwise comparisons with multiple-comparison correction), the claim that the effective range is "stable" is not fully supported.

Finally, the skill composition results (Table \ref{tab:compose}) are derived from a very small sample of episodes (31 total). While the percentage differences are large, the statistical power is low. The authors should provide confidence intervals for these proportions or apply appropriate statistical tests (e.g., Fisher's exact test) to confirm that the gains from Component Merging are not due to chance. The current presentation treats these percentages as deterministic facts rather than estimates with uncertainty.

Action items.

- **[science]** The sensitivity analysis (Appendix app:perturbation) reports single-point estimates for success rates under perturbations (e.g., 'Plaintext' on ALFWorld) without confidence intervals or standard deviations. Given the stochastic nature of LLM agent trajectories, a single run per condition is insufficient to establish statistical significance. Please report results averaged over multiple random seeds (e.g., 3-5) with standard errors or 95% confidence intervals to validate the robustness claims.
- **[science]** In the injection coefficient analysis (Appendix app:scale), the optimal alpha values are selected based on peak performance on a single run. The paper claims a 'stable effective injection range' but does not provide statistical tests (e.g., ANOVA or pairwise t-tests with correction) to confirm that the differences between alpha levels (e.g., 0.5 vs 0.6) are statistically

significant rather than noise. Please include significance testing for the scale-performance curves.

- [**science**] The skill composition results (Table tab:compose) are based on a small sample size (31 episodes total: 13 seen, 18 unseen). The reported improvements (e.g., 84.6% vs 61.5%) lack statistical validation. With such small N, the variance is high. Please report confidence intervals for these proportions or perform a statistical test (e.g., McNemar's test for paired episodes if applicable, or Fisher's exact test) to support the claim that Component Merging is significantly superior.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical sophistication, and the writing is generally clear, professional, and well-structured. The logical flow from the problem statement (context overhead) to the proposed solution (LatentSkill) and subsequent analysis is coherent. The abstract effectively summarizes the contributions, and the introduction sets the stage well.

However, there are a few minor grammatical inconsistencies and article usage errors that detract slightly from the overall polish. Specifically, in Section 4.3, the sentence "To examine whether \method{} form a meaningful latent geometry" exhibits a subject-verb agreement error; the singular subject requires the verb "forms." Similarly, in Section 4.4, the text inconsistently uses "a injection coefficient" instead of the grammatically correct "an injection coefficient." These are straightforward fixes but should be addressed to ensure the manuscript meets the highest standards of academic writing.

Additionally, while the figure captions are generally informative, the caption for Figure 1 could be slightly refined for absolute clarity regarding the "zero skill tokens" claim. The prose in the case studies (Appendix) is particularly strong, offering detailed and easy-to-follow explanations of failure modes. Overall, the writing quality is strong, and these minor issues do not impede understanding but warrant correction before final publication.

Action items.

- [**writing**] In Section 4.3 (Structured), the sentence 'To examine whether \method{} form a meaningful latent geometry' contains a subject-verb agreement error. The subject '\method{}' (referring to the framework or the resulting weights) is singular, but the verb 'form' is plural. It should be corrected to 'forms' or the subject rephrased to 'the LoRA weights generated by \method{} form' for clarity.

- [**writing**] In Section 4.4 (Controllable), the phrase 'We introduce an injection coefficient' uses the indefinite article 'an' before 'injection', which starts with a vowel sound. While grammatically acceptable, the subsequent sentence 'Specifically, we introduce a injection coefficient' incorrectly uses 'a' before 'injection'. This inconsistency and error should be fixed to 'an injection coefficient' throughout the section.

- **[writing]** In the caption of Figure 1 (motivation_1.png), the phrase 'zero skill tokens in prompt' is slightly ambiguous. It could be misinterpreted as 'zero skill tokens [are allowed] in [the] prompt'. Consider rephrasing to 'zero skill tokens in the prompt' or 'no skill tokens in the prompt' for better readability and precision.